NATIONAL MATHEMATICS CONTEST
A-LEVEL PAPER 2 SOLUTIONS.

SECTION A: 5 marks each

1. Solve the equation $(x^2+3x-4)^3+(2x^2-5x+3)^3=(3x^2-2x-1)^3$.

Solution 1 : *If*

$$x^2+3x-4=a, 2x^2-5x+3=b$$

then

$$3x^2-2x-1=a+b$$

So we have

$$a^3+b^3=(a+b)^3=a^3+3a^2b+3ab^2+b^3$$

or

$$3ab(a+b)=0$$

So we need to solve

$$x^2+3x-4=0\Rightarrow x=1,-4$$
$$2x^2-5x+3=0\Rightarrow x=1,\frac{3}{2}$$
$$3x^2-2x-1=0\Rightarrow x=1,-\frac{1}{3}$$

And the solution set $x\in\left\{-4,-\frac{1}{3},1,\frac{3}{2}\right\}$ ■

Solution 2 :

$$\left(x^2+3x-4\right)^3+\left(2x^2-5x+3\right)^3=\left(3x^2-2x-1\right)^3$$
$$(x-1)^3(x+4)^3+(x-1)^3(2x-3)^3-(x-1)^3(3x+1)^3=0$$
$$(x-1)^3\left((x+4)^3+(2x-3)^3-(3x+1)^3\right)=0$$
$$(x-1)^3\left((x+4)^3-(x+4)\left(19x^2-13x+7\right)\right)=0$$
$$(x-1)^3(x+4)\left(21x-18x^2+9\right)=0$$
$$x=\left\{-4,-\frac{1}{3},1,\frac{3}{2}\right\}$$

■

Solution 3 :

$$\left(x^2+3x-4\right)^3+\left(2x^2-5x+3\right)^3=\left(3x^2-2x-1\right)^3$$
$$\Leftrightarrow -18x^6+3x^5+192x^4-378x^3+222x^2+15x-36=0$$
$$\Leftrightarrow -3(x-1)^3(x+4)(2x-3)(3x+1)=0$$
$$\Leftrightarrow x\in\left\{-4,-\frac{1}{3},1,\frac{3}{2}\right\}$$

■

Solution 4 : $\Rightarrow 9x^6 - 51x^5 + 201x^4 - 350x^3 + 219x^2 + 9x - 37 = 27x^6 - 54x^5 + 9x^4 + 28x^3 - 3x^2 - 6x - 1 \Rightarrow -18x^6 + 3x^5 + 192x^4 - 378x^3 + 222x^2 + 15x - 36 = 0$ $\Rightarrow -3(x-1)^3(x+4)(2x-3)(3x+1) = 0$ *after IVT and RRT So* $x \in \left\{-4, -\frac{1}{3}, 1, \frac{3}{2}\right\}$.

■

2. Prove that for all positive real numbers a, b and c, $\dfrac{a^3}{bc} + \dfrac{b^3}{ca} + \dfrac{c^3}{ab} \geq a + b + c$.

Solution 1 : *Note that*

$$a^4 + b^4 + c^4 = \frac{(a^4 + b^4)}{2} + \frac{(b^4 + c^4)}{2} + \frac{(c^4 + a^4)}{2}.$$

Applying the arithmetic-geometric mean inequality to each term, we see that the right side is greater than or equal to

$$a^2b^2 + b^2c^2 + c^2a^2.$$

We can rewrite this as

$$\frac{a^2(b^2 + c^2)}{2} + \frac{b^2(c^2 + a^2)}{2} + \frac{c^2(a^2 + b^2)}{2}.$$

Applying the arithmetic mean-geometric mean inequality again we obtain

$$a^4 + b^4 + c^4 \geq a^2bc + b^2ca + c^2ab.$$

Dividing both sides by abc (which is positive) the result follows. ■

Solution 2 : *Notice the inequality is homogeneous. That is, if a, b, c are replaced by $ka, kb, kc, k > 0$ we get the original inequality. Thus we can assume, without loss of generality, that $abc = 1$.*
Then

$$\begin{aligned} \frac{a^3}{bc} + \frac{b^3}{ca} + \frac{c^3}{ab} &= abc\left(\frac{a^3}{bc} + \frac{b^3}{ca} + \frac{c^3}{ab}\right) \\ &= a^4 + b^4 + c^4. \end{aligned}$$

So we need prove that $a^4 + b^4 + c^4 \geq a + b + c$.

By the Power Mean Inequality,

$$\frac{a^4 + b^4 + c^4}{3} \geq \left(\frac{a + b + c}{3}\right)^4,$$

so

$$a^4 + b^4 + c^4 \geq (a + b + c) \cdot \frac{(a + b + c)^3}{27}.$$

By the arithmetic mean-geometric mean inequality,

$$\frac{a + b + c}{3} \geq \sqrt[3]{abc} = 1,$$

so

$$a + b + c \geq 3.$$

Hence,

$$a^4 + b^4 + c^4 \geq (a + b + c) \cdot \frac{(a + b + c)^3}{27} \geq (a + b + c)\frac{3^3}{27} = a + b + c.$$

■

Solution 3 : *Rather than using the Power-Mean inequality to prove* $a^4+b^4+c^4 \geq a+b+c$ *in Proof 2, the Cauchy-Schwartz-Bunjakovsky inequality can be used twice.*

$$(a^4+b^4+c^4)(1^2+1^2+1^2) \geq (a^2+b^2+c^2)^2$$

$$(a^2+b^2+c^2)(1^2+1^2+1^2) \geq (a+b+c)^2$$

So

$$\frac{a^4+b^4+c^4}{3} \geq \frac{(a^2+b^2+c^2)^2}{9} \geq \frac{(a+b+c)^4}{81}$$

Continue as in Solution 2. ■

3. There are 20 students in a high school class, and each student has exactly three close friends in the class. Five of the students have bought tickets to an upcoming concert. If any student sees that at least two of their close friends have bought tickets, then they will buy a ticket too. Is it possible that the entire class buys tickets to the concert? (Assume that friendship is mutual; if student A is close friends with student B, then B is close friends with A).

Solution 1 : *It is impossible for the whole class to buy tickets to the concert. If two students* A *and* B *are close friends, and* A *has bought a ticket to the concert while* B *has not, then* A *is enticing* B*. We call this pair* (A, B) *an enticement.*

In order for a student to change their mind and buy a ticket, they first be enticed by at least 2 *of their* 3 *close friends. That means they can only entice at most* 1 *other friend. Therefore, the total number of enticements among the students decreases by* 1 *whenever a student changes their mind to buy a ticket.*

Initially, the maximum number of enticements is 15 *(each of the initial* 5 *students with tickets has* 3 *friends to entice). Assume, for the sake of contradiction, that the entire class ends up buying tickets. After the first* 14 *people buy tickets, the number of enticements is at most* $15-14=1$*. This is not enough to convince the last person to buy a ticket, since they need* 2 *enticements.*

Therefore, it is impossible that the entire class buys tickets. ■

Solution 2 : *We shall use the term* friendship *to denote an unordered pair of students who are close friends. Since each of the* 20 *students is part of exactly* 3 *friendships, there are exactly* 30 *friendships in the class. (We could also represent friendships as edges in an undirected graph whose vertices are the* 20 *students.)*

We say that a friendship is used *if one of the students in that friendship buys a ticket after the original five buyers, and the other student already has a ticket at that time. Each time a ticket is purchased after the original five purchases, at least two friendships are used. Observe that no friendship gets used twice.*

If all 20 *students buy tickets, then three friendships are used when the last student buys a ticket. This would imply that the number of used friendships is at least* $14\times 2+3=31$*, which is more than the number of friendships. This contradiction proves that it is not possible that the entire class buys tickets.* ■

4. The circumradius R of a triangle with side lengths a, b, c satisfies $R = \dfrac{a\sqrt{bc}}{b+c}$. Find the angles of the triangle.

Solution 1 : *since area* $= \dfrac{abc}{4R} = \dfrac{\sqrt{bc}(b+c)}{4} = \dfrac{1}{2}bc\sin A \leq \dfrac{bc}{2}$ *we obtain* $\dfrac{b+c}{2} \leq \sqrt{bc}$ *It follows* $\dfrac{\sqrt{bc}(b+c)}{4} \leq \dfrac{bc}{2}$ *hence* $b = c$ *and* $\sin A = 1 \Longrightarrow \angle A = 90^\circ$ *so* $\angle B = \angle C = 45^\circ$ ■

Solution 2 : *From sine rule* $\dfrac{a}{2\sin A} = R$, *substituting we get* $b + c = 2\sin A\sqrt{bc}$ *But from* $AM - GM$ *we have* $b + c \geq 2\sqrt{bc}$, *hence* $\sin A = 1$ *or* $A = 90^o$ *and moreover* $b = c$, *giving* $B = C = 45^o$ ■

5. Let $a > 0$ and x_1, x_2, x_3 be real numbers with $x_1 + x_2 + x_3 = 0$. Prove that;

$$\log_2(1 + a^{x_1}) + \log_2(1 + a^{x_2}) + \log_2(1 + a^{x_3}) \geq 3.$$

Solution : *Use the property of the log function to get that*

$$LHS = log_2\left(a^{x_1+x_2+x_3} + a^{x_1+x_2} + a^{x_3+x_2} + a^{x_1+x_3} + a^{x_1} + a^{x_2} + a^{x_3} + 1\right).$$

Use that $x_1 + x_2 + x_3 = 0$, *this simplifies to*

$$LHS = 2 + \left(a_{x_1} + \frac{1}{a^{x_1}}\right) + \left(a_{x_2} + \frac{1}{a^{x_2}}\right) + \left(a_{x_3} + \frac{1}{a^{x_3}}\right).$$

Using AM-GM we get that this expression is greater than or equal to

$$2 + 2 + 2 + 2 = 8.$$

So the $LHS \geq 8$. *Take log on both sides to get that*

$$log_2(LHS) \geq log_2(8) = 3.$$

■

6. Find all non-negative integers x, y such that $x^3y + x + y = xy + 2xy^2$.

Solution 1 : *Case 1:* $xy = 0$. *In this case the other is also visibly zero, so we're done.*

Case 2: $xy \neq 0$. *Observe that we get* $x + y = xy(1 + 2y - x)$. *In particular,* $xy \mid x + y \Rightarrow xy \leq x + y \Rightarrow (x-1)(y-1) \leq 1$. *Now if* $(x-1)(y-1) = 0$, *we have* $x = 1$ *or* $y = 1$, *and in both cases one can check that the other is* 1 *as well. Otherwise,* $x - 1 = y - 1 = 1$(*It can't be* -1 *because that would mean* $xy = 0$). *In other words,* $x = y = 2$. *Since all the solutions* $(0,0), (1,1), (2,2)$ *work, we're done with the problem and these are the solutions.* ■

Solution 2 : *The only solutions are* $(0,0), (1,1), (2,2)$, *which obviously work.*

Taking the equation modulo xy, *we see that* xy *must divide* $x + y$. *Since* x, y *are non-negative, this also implies that* $xy \leq x + y$, *so* $(x-1)(y-1) \leq 1$.

If x, y *were both greater than* 1, *we would have* $x - 1$ *and* $y - 1$ *are positive integers multiplying to a number at most* 1, *meaning that* $x = y = 2$.

If one of x, y *were* 0, *then clearly the other one is zero also.*

If $x = 1$, *then* $y \mid y + 1 \Rightarrow y = 1$. *If* $y = 1$, *then* $x \mid x + 1 \Rightarrow x = 1$.

We have exhausted all cases, so we are done . ■

Solution 3 : *If* $xy \neq 0 \Rightarrow x \mid y$ *and* $y \mid x$ *so,* $x = y \Rightarrow x^4 + 2x = x^2 + 2x^3 \Rightarrow x^2(x^2-1) = 2x(x^2-1)$ *thus:* $\bullet\ x^2 - 1 = 0 \Rightarrow x = 1$ *Hence,* $(x, y) = (1, 1)$ $\bullet\ x^2 - 1 \neq 0 \Rightarrow x^2 = 2x \Rightarrow x = 2$ *Hence,* $(x, y) = (2, 2)$ *If* $xy = 0$ *obviously* $(x, y) = (0, 0)$ *Therefore,*

$$(x, y) = (0, 0); (1, 1); (2, 2)$$

■

7. Call a triple (a, b, c) of positive numbers mysterious if

$$\sqrt{a^2 + \frac{1}{a^2c^2} + 2ab} + \sqrt{b^2 + \frac{1}{b^2a^2} + 2bc} + \sqrt{c^2 + \frac{1}{c^2b^2} + 2ca} = 2(a + b + c).$$

Prove that if the triple (a, b, c) is mysterious, then so is the triple (c, b, a).

Solution : *If* $abc < 1$ *we have*

$$\begin{aligned}\sqrt{a^2 + \frac{1}{a^2c^2} + 2ab} + \sqrt{b^2 + \frac{1}{b^2a^2} + 2bc} + \sqrt{c^2 + \frac{1}{c^2b^2} + 2ca} \\ > \sqrt{a^2 + b^2 + 2ab} + \sqrt{b^2 + c^2 + 2bc} + \sqrt{c^2 + a^2 + 2ca} \\ = (a + b) + (b + c) + (c + a) = 2(a + b + c).\end{aligned}$$

Analogously

$$\sqrt{a^2 + \frac{1}{a^2c^2} + 2ab} + \sqrt{b^2 + \frac{1}{b^2a^2} + 2bc} + \sqrt{c^2 + \frac{1}{c^2b^2} + 2ca} = 2(a + b + c)$$

for $abc = 1$ *and*

$$\sqrt{a^2 + \frac{1}{a^2c^2} + 2ab} + \sqrt{b^2 + \frac{1}{b^2a^2} + 2bc} + \sqrt{c^2 + \frac{1}{c^2b^2} + 2ca} < 2(a + b + c)$$

for $abc > 1$.

So (a, b, c) *is mysterious iff* $abc = 1$. *In particular if* (a, b, c) *is mysterious, so is* (c, b, a).
■

8. Determine the number of real solutions a to the equation

$$\left\lfloor \frac{1}{2}a \right\rfloor + \left\lfloor \frac{1}{3}a \right\rfloor + \left\lfloor \frac{1}{5}a \right\rfloor = a$$

Here, if x is a real number, then $\lfloor x \rfloor$ denotes the greatest integer that is less than or equal to x.

Solution : *Let* $a = 30k + r$, *where* k *is an integer,* r *is a real number between* 0 *and* 29 *inclusive and* $lcm(2, 3, 5) = 30$.

Then $\left\lfloor \frac{1}{2}a \right\rfloor = \left\lfloor \frac{1}{2}(30k + r) \right\rfloor = 15k + \left\lfloor \frac{r}{2} \right\rfloor$.

Similarly $\left\lfloor \frac{1}{3}a \right\rfloor = 10k + \left\lfloor \frac{r}{3} \right\rfloor$ *and* $\left\lfloor \frac{1}{5}a \right\rfloor = 6k + \left\lfloor \frac{r}{5} \right\rfloor$.

Now, $\left\lfloor \frac{1}{2}a \right\rfloor + \left\lfloor \frac{1}{3}a \right\rfloor + \left\lfloor \frac{1}{5}a \right\rfloor = a$,

so $\left(15k + \left\lfloor \frac{r}{2} \right\rfloor\right) + \left(10k + \left\lfloor \frac{r}{3} \right\rfloor\right) + \left(6k + \left\lfloor \frac{r}{5} \right\rfloor\right) = 30k + r$

and hence $k = r - \left\lfloor \frac{r}{2} \right\rfloor - \left\lfloor \frac{r}{3} \right\rfloor - \left\lfloor \frac{r}{5} \right\rfloor$.

Clearly, r *has to be an integer, or* $r - \left\lfloor \frac{r}{2} \right\rfloor - \left\lfloor \frac{r}{3} \right\rfloor - \left\lfloor \frac{r}{5} \right\rfloor$ *will not be an integer, and therefore, cannot equal* k.

On the other hand, if r *is an integer, then* $r - \left\lfloor \frac{r}{2} \right\rfloor - \left\lfloor \frac{r}{3} \right\rfloor - \left\lfloor \frac{r}{5} \right\rfloor$ *will also be an integer, giving exactly one solution for* k.

For each $r (0 \leq r \leq 29), a = 30k + r$ *will have a different remainder* mod 30*, so no two different values of* r *give the same result for* a.

Since there are 30 *possible values for* $r (0, 1, 2, \ldots, 29)$*, there are then* 30 *solutions for* a. ■

9. Let S be a subset of $\{1, 2, \ldots, 9\}$, such that the sums formed by adding each ordered pair of distinct numbers from S are all different. For example, the subset $\{1, 2, 3, 5\}$ has this property, but $\{1, 2, 3, 4, 5\}$ does not, since the pairs $\{1, 4\}$ and $\{2, 3\}$ have the same sum, namely 5.

   What is the maximum number of elements that S can contain?

   Solution 1 : *It can be checked that all the sums of pairs for the set* $\{1, 2, 3, 5, 8\}$ *are different.*

   Suppose, for a contradiction, that S *is a subset of* $\{1, \ldots, 9\}$ *containing* 6 *elements such that all the sums of pairs are different. Now the smallest possible sum for two numbers from* S *is* $1 + 2 = 3$ *and the largest possible sum is* $8 + 9 = 17$*. That gives* 15 *possible sums:* $3, \ldots, 17$.

   Also there are $\binom{6}{2} = 15$ *pairs from* S*. Thus, each of* $3, \ldots, 17$ *is the sum of exactly one pair. The only pair from* $\{1, \ldots, 9\}$ *that adds to* 3 *is* $\{1, 2\}$ *and to* 17 *is* $\{8, 9\}$*. Thus* $1, 2, 8, 9$ *are in* S*. But then* $1 + 9 = 2 + 8$*, giving a contradiction. It follows that the maximum number of elements that* S *can contain is* 5. ■

   Solution 2 : *It can be checked that all the sums of pairs for the set* $\{1, 2, 3, 5, 8\}$ *are different.*

   Suppose, for a contradiction, that S *is a subset of* $\{1, \ldots 9\}$ *such that all the sums of pairs are different and that* $a_1 < a_2 < \ldots < a_6$ *are the members of* S.

   Since $a_1 + a_6 \neq a_2 + a_5$*, it follows that* $a_6 - a_5 \neq a_2 - a_1$.
   Similarly $a_6 - a_5 \neq a_4 - a_3$ *and* $a_4 - a_3 \neq a_2 - a_1$*. These three differences must be distinct positive integers, so,*

$$(a_6 - a_5) + (a_4 - a_3) + (a_2 - a_1) \geq 1 + 2 + 3 = 6.$$

   Similarly $a_3 - a_2 \neq a_5 - a_4$*, so*

$$(a_3 - a_2) + (a_5 - a_4) \geq 1 + 2 = 3.$$

   Adding the above 2 inequalities yields

$$a_6 - a_5 + a_5 - a_4 + a_4 - a_3 + a_3 - a_2 + a_2 - a_1 \geq 6 + 3 = 9,$$

   and hence $a_6 - a_1 \geq 9$*. This is impossible since the numbers in* S *are between* 1 *and* 9. ■

10. Let $ABCD$ be a convex quadrilateral with $\angle CBD = 2\angle ADB$, $\angle ABD = 2\angle CDB$ and $AB = CB$. Prove that $AD = CD$.

Solution 1 : *Extend DB to a point P on the circle through A and C centered at B. Then*

$$\angle CPD = \frac{1}{2}\angle CBD = \angle ADB$$

and

$$\angle APD = \frac{1}{2}\angle ABD = \angle CDB,$$

so $APCD$ is a parallelogram. Now PD bisects AC so BD is an angle bisector of isosceles triangle ABC. We have

$$\angle ADB = \frac{1}{2}\angle CBD = \frac{1}{2}\angle ABD = \angle CDB$$

so DB is the angle bisector of $\angle ADC$. As DB bisects the base of triangle ADC, this triangle must be isosceles and $AD = CD$. ■

Solution 2 : *Let the bisector of $\angle ABD$ meet AD at E. Let the bisector of $\angle CBD$ meet CD at F. Then*

$$\angle FBD = \angle BDE$$

and

$$\angle EBD = \angle BDF,$$

which imply $BE \parallel FD$ and $BF \parallel ED$. Thus $BEDF$ is a parallelogram whence

$$BD \text{ intersects } EF \text{ at its midpoint } M. \tag{1.1}$$

On the other hand since BE is an angle bisector, we have $\frac{AB}{BD} = \frac{AE}{ED}$. Similarly we have

$$\frac{CB}{BD} = \frac{CF}{FD}.$$

By assumption $AB = CB$ so

$$\frac{AE}{ED} = \frac{CF}{FD}$$

which implies $EF \parallel AC$. Thus $\triangle DEF$ and $\triangle DAC$ are similar, which implies by (1.1) that BD intersects AC at its midpoint N. Since $\triangle ABC$ is isosceles, this implies $AC \perp BD$. Thus $\triangle NAD$ and $\triangle NCD$ are right triangles with equal legs and hence are congruent. Thus $AD = CD$.

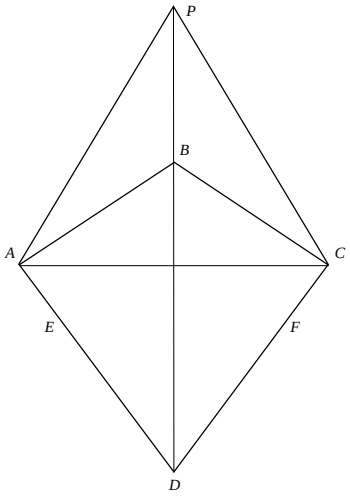


Diagram for Solution 1

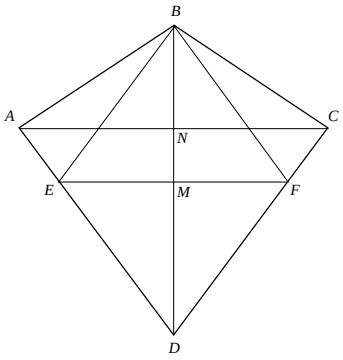


Diagram for Solution 2

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SECTION B: 10 marks each

11. Find all real numbers x such that $x = \left(x - \frac{1}{x}\right)^{1/2} + \left(1 - \frac{1}{x}\right)^{1/2}$.

Solution : *Since*

$$\left(x - \frac{1}{x}\right)^{1/2} \geq 0$$

and

$$\left(1 - \frac{1}{x}\right)^{1/2} \geq 0,$$

then

$$0 \leq \left(x - \frac{1}{x}\right)^{1/2} + \left(1 - \frac{1}{x}\right)^{1/2} = x.$$

Note that $x \neq 0$. *Else,* $\frac{1}{x}$ *would not be defined so* $x > 0$.

Squaring both sides gives,

$$\begin{aligned} x^2 &= \left(x - \frac{1}{x}\right) + \left(1 - \frac{1}{x}\right) + 2\sqrt{\left(x - \frac{1}{x}\right)\left(1 - \frac{1}{x}\right)} \\ x^2 &= x + 1 - \frac{2}{x} + 2\sqrt{x - 1 - \frac{1}{x} + \frac{1}{x^2}} \end{aligned}$$

Multiplying both sides by x and rearranging, we get

$$\begin{aligned} x^3 - x^2 - x + 2 &= 2\sqrt{x^3 - x^2 - x + 1} \\ (x^3 - x^2 - x + 1) - 2\sqrt{x^3 - x^2 - x + 1} + 1 &= 0 \\ (\sqrt{x^3 - x^2 - x + 1} - 1)^2 &= 0 \\ \sqrt{x^3 - x^2 - x + 1} &= 1 \\ x^3 - x^2 - x + 1 &= 1 \\ x(x^2 - x - 1) &= 0 \\ x^2 - x - 1 &= 0 \ \textit{since} \ \ x \neq 0 : \end{aligned}$$

Thus $x = \dfrac{1 \pm \sqrt{5}}{2}$. *We must check to see if these are indeed solutions.*

Let $\alpha = \dfrac{1 + \sqrt{5}}{2}, \beta = \dfrac{1 - \sqrt{5}}{2}$. *Note that* $\alpha + \beta = 1, \alpha\beta = -1$ *and* $\alpha > 0 > \beta$.

Since $\beta < 0, \beta$ *is not a solution.*

Now, if $x = \alpha$ *, then*

$$\begin{aligned} \left(\alpha - \frac{1}{\alpha}\right)^{1/2} + \left(1 - \frac{1}{\alpha}\right)^{1/2} &= (\alpha + \beta)^{1/2} + (1 + \beta)^{1/2} && (\textit{since} \quad \alpha\beta - -1) \\ &= 1^{1/2} + (\beta^2)^{1/2} && (\textit{since} \ \ \alpha + \beta = 1 \ \ \textit{and} \ \ \beta^2 = \beta + 1) \\ &= 1 - \beta && (\textit{since} \ \ \beta < 0) \\ &= \alpha && (\textit{since} \ \ \alpha + \beta = 1). \end{aligned}$$

So $x = \alpha$ *is the unique solution to the equation.* ■

12. (a) Find all prime numbers p and q such that $p^3 - 3^q = 10$.

Solution 1 : *If* $p \neq 13$, *we have*

$$p^3 \in \{1, 5, 8, 12\} \pmod{13}$$

and

$$3^q \in \{1, 3, 9\} \pmod{13}$$

$$\Rightarrow 10 = p^3 - 3^q \in \{0, 2, 3, 4, 5, 7, 9, 11, 12\} \pmod{13}, \ \textit{a contradiction.}$$

Therefore, $p = 13 \implies 3^q = 13^3 - 10 \implies q = 7$. *Thus,* $(p, q) = (13, 7)$ *is the only solution.* ■

Solution 2 : *Rearrange to get* $3^q + 10 = p^3$. *Notice that the residues* 3^x *give* mod 13 *are* $1, 3, 9$ *which means the possible* mod 13 *values for* p^3 *are* $11, 0, 6$ *while the cubic residues mod 13 possible are* $0, 1, 5, 7, 8, 9, 12$, *which means* $p^3 \equiv 0 \mod 13$, *which means* $p = 13$. *Substituting this back, we have* $q = 7$ ■

(b) Let a and b be real numbers such that

$$\frac{a}{a^2-5} = \frac{b}{5-b^2} = \frac{ab}{a^2b^2-5}$$

where $a+b \neq 0$. Determine the value of a^4+b^4.

Solution 1 : *By cross multiplication, we have*

$$5a - ab^2 = a^2b - 5b \implies 5(a+b) = ab(a+b) \implies ab = 5$$

Hence,

$$\frac{a}{a^2-5} = \frac{b}{5-b^2} = \frac{1}{4}$$

Again

$$\frac{a}{a^2-5} = \frac{1}{4} \implies a^2 - 4a - 5 = 0 \implies a \in \{-1, 5\}$$

Similarly, $b \in \{1, -5\}$. *Therefore,* $(a,b) = (1,5)$ *or* $(-1,-5)$ *or* $(5,1)$ *or* $(-5,-1)$ *as* $ab = 5$. *Which implies*

$$a^4 + b^4 = 626$$

■

Solution 2 : *Replacing* b *with* $-b$ *we got* $\frac{a}{a^2-5} = \frac{b}{b^2-5} = -\frac{ab}{a^2b^2-5}$ *ans easy to see, that* $ab = -5$

$$a + b = \frac{a^2-b^2}{a-b} = \frac{a^2-5}{a} = 4$$

$$a^2 + b^2 = (a+b)^2 - 2ab = 26$$

$$a^4 + b^4 = (a^2+b^2) - 2a^2b^2 = 626$$

■

Solution 3 : *Let* $\frac{1}{c}$ *the common value. We have three equations :*

$$\begin{aligned} a^2 - ac - 5 &= 0 \\ b^2 + bc - 5 &= 0 \\ (ab)^2 - (ab)c - 5 &= 0 \end{aligned}$$

First and third imply that a *and* ab *both are roots of* $X^2 - cX - 5 = 0$ *and so : : Either* $b = 1$ *(same root), either* $a(ab) = -5$ *and so* $b = -\frac{5}{a^2}$

Subtracting first and second, we get $(a+b)(a-b-c) = 0$ *and so, since* $a+b \neq 0$ *:* $b = a - c = \frac{5}{a}$ *(from first)*

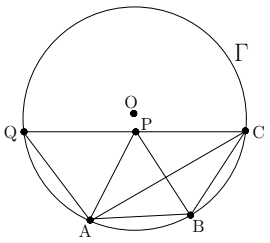
And so two cases :

1) $b = 1$ *and* $b = \frac{5}{a}$ *and so* $(a,b) = (5,1)$ *which indeed fits*

2) $b = -\frac{5}{a^2}$ *and* $b = \frac{5}{a}$ *and so* $(a,b) = (-1,-5)$, *which indeed fits*

And in both cases : $a^4 + b^4 = 626$ ■

13. (a) Let Γ be a circle with radius r. Let A and B be distinct points on Γ such that $AB < \sqrt{3}r$. Let the circle with centre B and radius AB meet Γ again at C. Let P be the point inside Γ such that triangle ABP is equilateral. Finally, let CP meet Γ again at Q. Prove that $PQ = r$.



Solution 1 :

Let the center of Γ *be* O*, the radius* r*. Since* $BP = BC$*, let*

$$\theta = \measuredangle BPC = \measuredangle BCP.$$

Quadrilateral $QABC$ *is cyclic, so* $\measuredangle BAQ = 180^\circ - \theta$ *and hence*

$$\measuredangle PAQ = 120^\circ - \theta.$$

Also $\measuredangle APQ = 180^\circ - \measuredangle APB - \measuredangle BPC = 120^\circ - \theta$ *, so*

$$PQ = AQ and \measuredangle AQP = 2\theta - 60^\circ.$$

Again because quadrilateral $QABC$ *is cyclic,*

$$\measuredangle ABC = 180^\circ - \measuredangle AQC = 240^\circ - 2\theta.$$

Triangles OAB *and* OCB *are congruent, since* $OA = OB = OC = r$ *and* $AB = BC$*.*

Thus

$$\measuredangle ABO = \measuredangle CBO = \frac{1}{2}\measuredangle ABC = 120^\circ - \theta.$$

We have now shown that in triangles AQP *and*

$$AOB, \measuredangle PAQ = \measuredangle BAO = \measuredangle APQ = \measuredangle ABO.$$

Also $AP = AB$*, so* $\triangle AQP \cong \triangle AOB$*. Hence* $QP = OB = r$*.*

■

Solution 2 :

Let the center of Γ *be* O*, the radius* r*. Since* A, P *and* C *lie on a circle centered at* $B, 60^\circ = \measuredangle ABP = 2\measuredangle ACP$*, so*

$$\measuredangle ACP = \measuredangle ACQ = 30^\circ.$$

Since $Q, A,$ *and* C *lie on*

$$\Gamma, \measuredangle QOA = 2\measuredangle QCA = 60^\circ$$

So $QA = r$ *since if a chord of a circle subtends an angle of* 60° *at the center, its length is the radius of the circle.*

Now $BP = BC$*, so*

$$\measuredangle BPC = \measuredangle BCP = \measuredangle ACB + 30^\circ.$$

Thus

$$\measuredangle APQ = 180^\circ - \measuredangle APB - \measuredangle BPC = 90^\circ - \measuredangle ACB$$

Since Q, A, B and C lie on Γ and

$$AB = BC, \measuredangle AQP = \measuredangle AQC = \measuredangle AQB + \measuredangle BQC = 2\measuredangle ACB$$

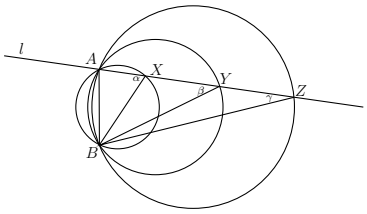
Finally,

$$\measuredangle QAP = 180 - \measuredangle AQP - \measuredangle APQ = 90 - \measuredangle ACB$$

So $\measuredangle PAQ = \measuredangle APQ$ hence $PQ = AQ = r$.

■

(b) Prove that when three circles share the same chord AB, every line through A different from AB determines the same ratio $XY : YZ$, where X is an arbitrary point different from B on the first circle while Y and Z are the points where AX intersects the other two circles (labelled so that Y is between X and Z).



Solution 1 : *Let l be a line through A different from AB and join B to A, X, Y and Z as in the above diagram. No matter how l is chosen, the angles AXB, AYB and AZB always subtend the chord AB. For this reason the angles in the triangles BXY and BXZ are the same for all such l. Thus the ratio $XY : YZ$ remains constant by similar triangles.*

Note that this is true no matter how X, Y and Z lie in relation to A. Suppose X, Y and Z all lie on the same side of A (as in the diagram) and that $\measuredangle AXB = \alpha, \measuredangle AYB = \beta$ and $\measuredangle AZB = \gamma$. Then

$$\measuredangle BXY = 180^\circ - \alpha, \measuredangle BYX = \beta, \measuredangle BYZ = 180^\circ - \beta$$

and $\measuredangle BZY = \gamma$. Now suppose l is chosen so that X is now on the opposite side of A from Y and Z. Now since X is on the other side of the chord $AB, \measuredangle AXB = 180^\circ - \alpha$, but it is still the case that $\measuredangle BXY = 180^\circ - \alpha$ and all other angles in the two pertinent triangles remain unchanged. If l is chosen so that X is identical with A, then l is tangent to the first circle and it is still the case that $\measuredangle BXY = 180^\circ - \alpha$. All other cases can be checked in a similar manner.

■

Solution 2 : *Let m be the perpendicular bisector of AB and let O_1, O_2, O_3 be the centres of the three circles. Since AB is a chord common to all three circles, O_1, O_2, O_3 all lie on m.*

Let l be a line through A different from AB and suppose that X, Y, Z all lie on the same side of AB, as in the above diagram. Let perpendiculars from O_1, O_2, O_3 meet l at P, Q, R, respectively. Since a line through the centre of a circle bisects any chord, $AX = 2AP, AY = 2AQ$ and $AZ = 2AR$. Now

$$XY = AY - AX = 2(AQ - AP) = 2PQ$$

and, similarly, $YZ = 2QR$ Therefore $XY : YZ = PQ : QR$. But $O_1P || O_2Q || O_3R$, so

$$PQ : QR = O_1O_2 : O_2O_3$$

Since the centres of the circles are fixed, the ratio $XY : YZ = O_1O_2 : O_2O_3$ does not depend on the choice of l.

If X, Y, Z do not all lie on the same side of AB, we can obtain the same result with a similar proof. For instance, if X and Y are opposite sides of AB, then we will have $XY = AY + AX$, but since in this case $PQ = AQ + AP$, it is still the case that $XY = 2PQ$ and result still follows, etc. ■

14. William is thinking of an integer between 1 and 50, inclusive. Victor can choose a positive integer m and ask William: "does m divide your number?", to which William must answer truthfully. Victor continues asking these questions until he determines William's number. What is the minimum number of questions that Victor needs to guarantee this?

Solution : *The minimum number is 15 questions. First, we show that 14 or fewer questions is not enough to guarantee success. Suppose Victor asks at most 14 questions, and William responds with "no" to each question unless $m = 1$. Note that these responses are consistent with the secret number being 1. But since there are 15 primes less than 50, some prime p was never chosen as m. That means the responses are also consistent with the secret number being p. Therefore, Victor cannot determine the number for sure because 1 and p are both possible options.*

Now we show that Victor can always determine the number with 15 questions.
Let N be William's secret number. First, Victor asks 4 questions, with $m = 2, 3, 5, 7$. We then case on William's responses.

Case 1. William answers "no" to all four questions.

N can only be divisible by primes that are 11 or larger. This means N cannot have multiple prime factors (otherwise $N \geq 112 > 50$), so either $N = 1$ or N is one of the 11 remaining primes less than 50. Victor can then ask 11 questions with $m = 11, 13, 17, ..., 47$, one for each of the remaining primes, to determine the value of N.

Case 2. William answers "yes" to $m = 2$, and "no" to $m = 3, 5, 7$.

There are only 11 possible values of N that match these answers ($2, 4, 8, 16, 22, 26, 32, 34, 38, 44$, and 46). Victor can use his remaining 11 questions on each of these possibilities.

Case 3. William answers "yes" to $m = 3$, and "no" to $m = 2, 5, 7$.

There are 5 possible values of N ($3, 9, 27, 33$, and 39). Similar to Case 142, Victor can ask about these 5 numbers to determine the value of N.

Case 4. William answers "yes" to multiple questions, or one "yes" to $m = 5$ or $m = 7$.

Let k be the product of all m's that received a "yes" response. Since N is divisible by each of these m's, N must be divisible by k. Since $k \geq 5$, there are at most 10 multiples of k between 1 and 50. Victor can ask about each of these multiples of k with his remaining questions.

■

15. (a) Let x, y and z be positive real numbers. Show that $x^2 + xy^2 + xyz^2 \geq 4xyz - 4$.

Solution : *Note that*

$$x^2 \geq 4x - 4, y^2 \geq 4y - 4, \quad \text{and} \quad z^2 \geq 4z - 4,$$

and therefore

$$x^2 + xy^2 + xyz^2 \geq (4x - 4) + x(4y - 4) + xy(4z - 4) = 4xyz - 4.$$

■

(b) The product of the positive real numbers x, y, z is 1. Show that if $\frac{1}{x}+\frac{1}{y}+\frac{1}{z} \geq x+y+z$, then $\frac{1}{x^k}+\frac{1}{y^k}+\frac{1}{z^k} \geq x^k+y^k+z^k$ for all positive integers k.

Solution 1 :

$$\frac{1}{x}+\frac{1}{y}+\frac{1}{z} \geq x+y+z \Leftrightarrow xy+xz+yz-x-y-z \geq 0 \Leftrightarrow (1-x)(1-y)(1-z) \geq 0.$$

If $1-x \geq 0$, $1-y \geq 0$, $1-z \geq 0$ *then* $\{x, y, z\} \subset (0, 1]$. *Hence,* $\{x^k, y^k, z^k\} \subset (0, 1]$. *Hence,* $(1-x^k)(1-y^k)(1-z^k) \geq 0$. *That is,*

$$\frac{1}{x^k}+\frac{1}{y^k}+\frac{1}{z^k} \geq x^k+y^k+z^k.$$

If $1-x \leq 0$, $1-y \leq 0$ *and* $1-z \geq 0$ *then* $1-x^k \leq 0$, $1-y^k \leq 0$, $1-z^k \geq 0$. *That is,*

$$(1-x^k)(1-y^k)(1-z^k) \geq 0$$

and

$$\frac{1}{x^k}+\frac{1}{y^k}+\frac{1}{z^k} \geq x^k+y^k+z^k.$$

■

Solution 2 : *Consider the function :* $f(t) = (t-x)(t-y)(t-z)$. *Then* $f(1) = (1-x)(1-y)(1-z) \geq 0$. *Since* x, y, z *are positive, we have* $s(1-x^{2k}) = s(1-x^k) = s(1-x)$ *(where* $s(n)$ *denotes the sign of* n*), so*

$$(1-x^{2k})(1-y^{2k})(1-z^{2k}) \geq 0$$

Dividing both sides by $(xyz)^k$ *we'll get the desired result from AM-GM inequality.* ■